Zixun Zhou

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EDUCATION

Goldsmiths, University of London Sep 2026 – present
MSc Virtual & Extended Realities

Royal College of Art Sep 2025 – Jul 2026
Graduate Diploma, Design Pathway
Completed Units 2–4 with two hardware-software prototypes bridging haptics, environmental sensing and interactive design. Targeting IDE (Innovation Design Engineering) programme.

University of Manchester Sep 2021 – Jun 2024
BSc Computer Science (Applied)
Software Engineering, Data Science, AI & Machine Learning, Operating Systems, Computer Architecture.

UAL Camberwell College of Arts Sep 2017 – Jun 2018
Foundation Diploma — Illustration & Game Design

PROJECTS

Anima — Ecological Emotional Translation Device RCA U3–U4, Jun 2026

- Built an interactive device that reads 8 environmental sensors (temperature, pH, light, moisture, VOC, sound, EC, humidity) via Arduino, processes them through a CfC neural network (PyTorch + NCP), and outputs real-time emotional states as a breathing WS2812 LED and a mobile web dashboard.
- Designed a "genome" system: three habitat presets (freshwater, indoor air, tomato soil) give the device distinct personalities — the same sensor reading produces different emotional responses depending on which ecosystem it represents.
- Developed a narrative engine that writes a first-person "life diary" from the organism's perspective, triggered by comfort-zone transitions rather than raw data ticks.
- Full-stack: Arduino firmware (C++), Python backend (Flask, PyTorch, serial I/O), front-end PWA, hardware assembly.

Chladni Skin — Tactile Neurofeedback Device RCA U2, Mar 2026

- Designed a haptic neurofeedback device for ADHD: a silicone membrane filled with iron powder over an electromagnetic coil array, driven by Arduino.
- Engineered three adaptive tactile modes — Focus (20 Hz pulse on palm acupuncture points), Relax (10 s sine-wave sweep across four motors), Alert (3-burst all-motor activation) — switchable over serial.
- Confirmed a literature gap: no prior research used deformation-based (non-vibration) haptics for neurofeedback (Fleury et al., 2020).
- Hardware: Arduino, 3 electromagnets (12 V), 4 vibration motors (3.3 V).

EXPERIENCE

UX Data Analyst Intern — WPP / VMLY&R Jun – Aug 2023

- Decomposed 8 analytical modules (80+ metrics) covering user value benchmarking, ride experience, customer segmentation, UK geographic distribution and competitive analysis.
- Produced data visualisations (geo-heatmaps, scatter plots, dashboards) and designed core presentation decks for client pitches.

SKILLS

Languages	Python, C/C++ (Arduino), Java, JavaScript, HTML/CSS, SQL
Frameworks	PyTorch, Flask, Spring Boot
Hardware	Arduino, serial protocols, sensor integration, WS2812 LED, electromagnetic actuators, 3D printing (FDM/SLA)
Design	Figma, Procreate, Photoshop, physical prototyping, silicone casting, PVC prop fabrication
Languages	Mandarin (native), English (fluent)